

Competency list.

The cell, transport and signaling

Name _____

Date _____

These are the twenty five things you have to be able to do by Sunday. Every exam question, lab and case in Week 2 comes from this list. Tick a box only when it is true.

HOW TO USE THE THREE BOXES

Read. You have read it, in the book or the notes, and it made sense at the time.

Drew it. You have drawn or worked it with the notes open beside you.

From memory. You did it on blank paper with nothing open. This is the only box that predicts an exam score, and it is the one to chase. A row with the first two ticked and the third empty is exactly where your study time should go.

Cell anatomy and tissues STRUCTURE

4 competencies

1	Body fluid compartments State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
2	Compartment separation and clinical volume shifts Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
3	Membrane composition and fluidity Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
4	Determinants of permeability Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.	☐ READ	☐ DREW IT	☐ FROM MEMOR'

Cell physiology and transport WHAT CROSSES THE MEMBRANE

12 competencies

5	Simple diffusion and Fick's law State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
6	Osmolarity and tonicity Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
7	Osmosis and cell volume Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.	☐ READ	☐ DREW IT	☐ FROM MEMOR'

8	Diffusion and osmosis experiment Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
9	Tonicity and red blood cells Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
10	Facilitated diffusion Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
11	Primary active transport Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
12	Secondary active transport Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
13	Transport maximum and saturation Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
14	Vesicular transport Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
15	Transepithelial transport Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
16	Transport simulation Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.	☐ READ	☐ DREW IT	☐ FROM MEMOR'

Cell signaling HOW CELLS TALK TO EACH OTHER

9 competencies

17	Signal types and range Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
18	Receptor location and ligand solubility Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.	☐ READ	☐ DREW IT	☐ FROM MEMOR'
19	G protein coupled receptors Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.	☐ READ	☐ DREW IT	☐ FROM MEMOR'

20 Second messengers

Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

☐ READ ☐ DREW IT ☐ FROM MEMOR'

21 Catalytic receptors and intracellular receptors

Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

☐ READ ☐ DREW IT ☐ FROM MEMOR'

22 Signal amplification

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

☐ READ ☐ DREW IT ☐ FROM MEMOR'

23 Receptor modulation

Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

☐ READ ☐ DREW IT ☐ FROM MEMOR'

24 Signal termination

Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

☐ READ ☐ DREW IT ☐ FROM MEMOR'

25 Dose response relationships

Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

☐ READ ☐ DREW IT ☐ FROM MEMOR'